

KPI ADVISORY & ANALYTICS

Functional Specification Document

A unified blueprint translating business strategy into governed, measurable performance metrics.

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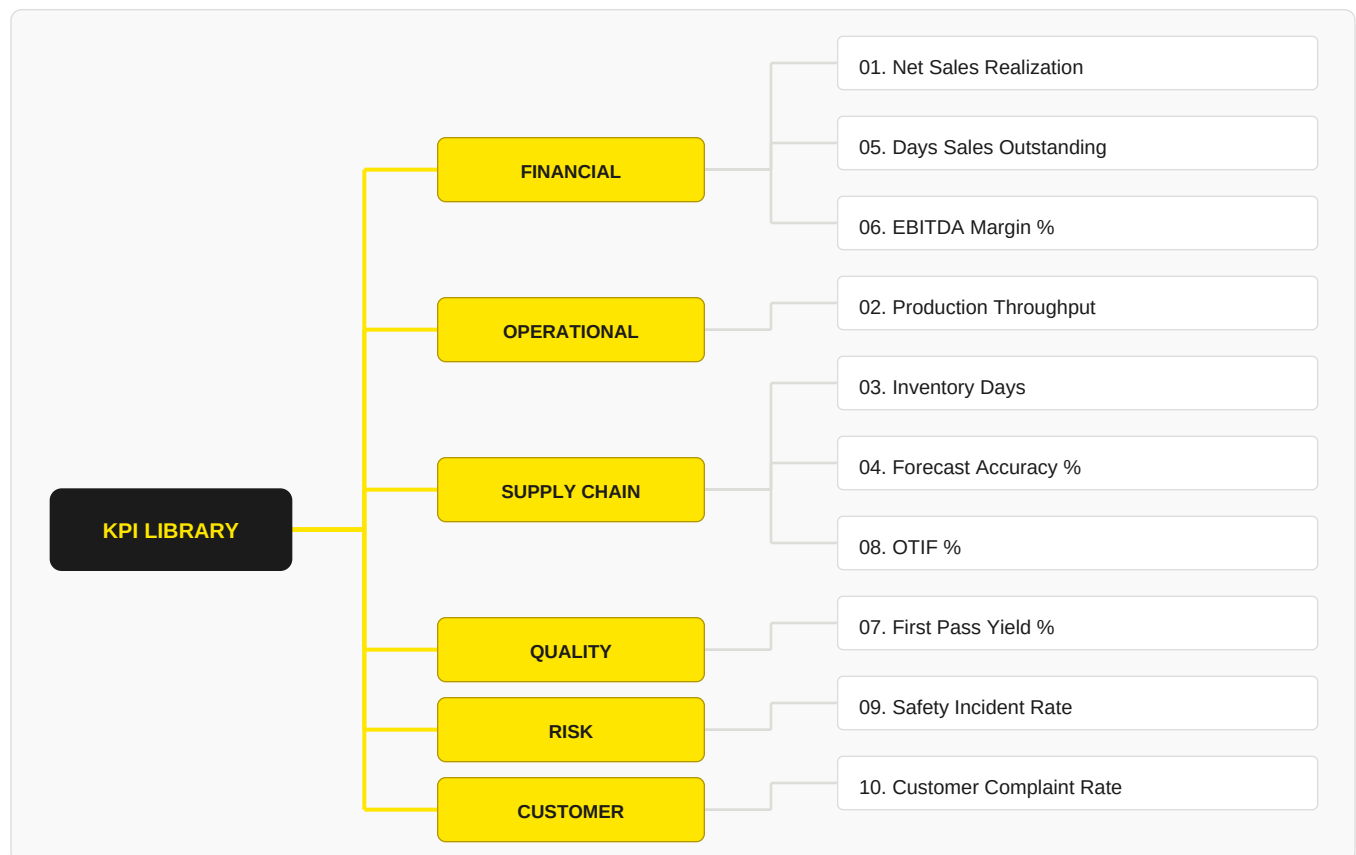
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1. Executive Summary

This document outlines the functional specifications for the approved performance metrics of the organization. Each metric is mapped to strategic objectives and key result areas to ensure operational alignment and governance.

2. KPI Landscape Overview

The following visual mind-map displays the KPI Landscape as a branching tree structure, routing from the core KPI Library through strategic categories down to individual approved metrics.



The following table provides a high-level catalog of all approved performance indicators within the scope of this transformation initiative.

KPI ID	KPI Name	Category	Functional Area
KPI-001	Net Sales Realization	Financial	Sales
KPI-002	Production Throughput	Operational	Production
KPI-003	Inventory Days	Supply Chain	Supply Chain

KPI-004	Forecast Accuracy %	Supply Chain	Supply Chain
KPI-005	Days Sales Outstanding	Financial	Finance
KPI-006	EBITDA Margin %	Financial	Finance
KPI-007	First Pass Yield %	Quality	Production
KPI-008	OTIF %	Supply Chain	Supply Chain
KPI-009	Safety Incident Rate	Risk	Operations
KPI-010	Customer Complaint Rate	Customer	Quality

3. Strategic Traceability Matrix

This matrix illustrates the strategic alignment from executive objectives down to specific key performance indicators, providing visibility into strategic translation.

Strategic Objective	Business Challenge	KRA	Functional Area	KPI Name
Improve Cash Flow	Margin Leakage from Discounting	Profitability	Sales	Net Sales Realization
Improve Cash Flow	Margin Leakage from Discounting	Operational Excellence	Production	Production Throughput
Improve Cash Flow	Margin Leakage from Discounting	Asset Productivity	Supply Chain	Inventory Days
Improve Cash Flow	Margin Leakage from Discounting	Operational Excellence	Supply Chain	Forecast Accuracy %
Improve Cash Flow	Margin Leakage from Discounting	Cash Flow	Finance	Days Sales Outstanding
Improve Cash Flow	Margin Leakage from Discounting	Profitability	Finance	EBITDA Margin %
Improve Cash Flow	Margin Leakage from Discounting	Operational Excellence	Production	First Pass Yield %
Improve Cash Flow	Margin Leakage from Discounting	Customer Growth	Supply Chain	OTIF %
Improve Cash Flow	Margin Leakage from Discounting	Risk Management	Operations	Safety Incident Rate
Improve Cash Flow	Margin Leakage from Discounting	Customer Growth	Quality	Customer Complaint Rate

4. Individual KPI Specifications

Metric 01: Net Sales Realization

KPI Category	Financial	Functional Area	Sales
Business Owner	VP of Sales	Data Owner	Sales Operations Manager

Business Definition

Measures and evaluates the efficiency of Net Sales Realization against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Profitability. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	Gross Sales Revenue - (Returns + Discounts + Rebates + Promos)
Numerator Details	Gross Sales Revenue - (Returns + Discounts + Rebates + Promos)
Denominator Details	Gross Sales Revenue
Calculation Logic	1. Extract the raw transaction history from SAP S/4HANA Sales. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP S/4HANA Sales (Module: SD). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 02: Production Throughput

KPI Category	Operational	Functional Area	Production
Business Owner	Head of Manufacturing	Data Owner	Production Planning Manager

Business Definition

Measures and evaluates the efficiency of Production Throughput against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Operational Excellence. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	Total Units Produced / Total Production Hours
Numerator Details	Total Units Produced
Denominator Details	Total Production Hours
Calculation Logic	1. Extract the raw transaction history from SAP MES. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP MES (Module: PP). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 03: Inventory Days

KPI Category	Supply Chain	Functional Area	Supply Chain
Business Owner	Head of Supply Chain	Data Owner	Inventory Control Manager

Business Definition

Measures and evaluates the efficiency of Inventory Days against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Asset Productivity. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Average Inventory Value} / \text{Annual Cost of Goods Sold}) * 365$
Numerator Details	Average Inventory Value
Denominator Details	Annual Cost of Goods Sold (COGS)
Calculation Logic	1. Extract the raw transaction history from SAP MM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP MM (Module: MM). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 04: Forecast Accuracy %

KPI Category	Supply Chain	Functional Area	Supply Chain
Business Owner	Head of Demand Planning	Data Owner	Demand Planning Analyst

Business Definition

Measures and evaluates the efficiency of Forecast Accuracy % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Operational Excellence. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	100 - Mean Absolute Percentage Error (MAPE)
Numerator Details	100 - (Sum of Absolute Errors / Total Actual Units)
Denominator Details	Total Actual Units
Calculation Logic	1. Extract the raw transaction history from SAP IBP (Integrated Business Planning). 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP IBP (Integrated Business Planning) (Module: MM/SD). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 05: Days Sales Outstanding

KPI Category	Financial	Functional Area	Finance
Business Owner	CFO / Head of Finance	Data Owner	Accounts Receivable Manager

Business Definition

Measures and evaluates the efficiency of Days Sales Outstanding against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Cash Flow. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Average Accounts Receivable} / \text{Total Credit Sales}) * 365$
Numerator Details	Average Accounts Receivable Balance
Denominator Details	Total Credit Sales in Period
Calculation Logic	1. Extract the raw transaction history from SAP FI-AR. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP FI-AR (Module: FI-AR). Dependencies: Master transactional tables and inventory logs.

Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 06: EBITDA Margin %

KPI Category	Financial	Functional Area	Finance
Business Owner	CFO / Head of Finance	Data Owner	Financial Controller

Business Definition

Measures and evaluates the efficiency of EBITDA Margin % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Profitability. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{EBITDA} / \text{Total Net Revenue}) * 100$
Numerator Details	Earnings before Interest, Tax, Depreciation, and Amortization
Denominator Details	Total Net Revenue
Calculation Logic	1. Extract the raw transaction history from SAP General Ledger. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP General Ledger (Module: FI). Dependencies: Master transactional tables and inventory logs.
Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 07: First Pass Yield %

KPI Category	Quality	Functional Area	Production
Business Owner	Head of Quality	Data Owner	Quality Assurance Manager

Business Definition

Measures and evaluates the efficiency of First Pass Yield % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Operational Excellence. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Good Units Produced with No Rework} / \text{Total Units Started}) * 100$
Numerator Details	Good Units Produced with No Rework
Denominator Details	Total Units Started
Calculation Logic	1. Extract the raw transaction history from SAP QM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP QM (Module: PP/QM). Dependencies: Master transactional tables and inventory logs.

Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 08: OTIF %

KPI Category	Supply Chain	Functional Area	Supply Chain
Business Owner	Head of Logistics	Data Owner	Logistics Coordinator

Business Definition

Measures and evaluates the efficiency of OTIF % against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Customer Growth. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Orders Delivered On-Time and In-Full} / \text{Total Orders Shipped}) * 100$
Numerator Details	Orders Delivered On-Time and In-Full
Denominator Details	Total Orders Shipped
Calculation Logic	1. Extract the raw transaction history from SAP SD / EWM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP SD / EWM (Module: SD/MM). Dependencies: Master transactional tables and inventory logs.
Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 09: Safety Incident Rate

KPI Category	Risk	Functional Area	Operations
Business Owner	Head of EHS (Environment, Health, and Safety)	Data Owner	EHS Manager

Business Definition

Measures and evaluates the efficiency of Safety Incident Rate against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Risk Management. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Number of Recordable Safety Incidents} * 200,000) / \text{Total Employee Hours Worked}$
Numerator Details	Number of Recordable Safety Incidents * 200,000
Denominator Details	Total Employee Hours Worked (Employees and Contractors)
Calculation Logic	1. Extract the raw transaction history from SAP EHS (Environment, Health, and Safety). 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP EHS (Environment, Health, and Safety) (Module: EHS). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

Metric 10: Customer Complaint Rate

KPI Category	Customer	Functional Area	Quality
Business Owner	Head of Customer Service	Data Owner	Customer Service Manager

Business Definition

Measures and evaluates the efficiency of Customer Complaint Rate against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Customer Growth. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	$(\text{Customer Complaints Received} / \text{Total Orders Shipped}) * 100$
Numerator Details	Customer Complaints Received
Denominator Details	Total Orders Shipped
Calculation Logic	1. Extract the raw transaction history from Salesforce CRM. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = $(9,500 / 10,000) * 100 = 95.00\%$.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: Salesforce CRM (Module: QM). Dependencies: Master transactional tables and inventory logs.
Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

5. Governance Framework

To ensure metric consistency, accountability, and ongoing relevance, a formal governance framework is established for all approved KPIs. This structure assigns clear responsibilities and defines escalation paths.

Roles and Responsibilities

- **Business Owner:** Responsible for defining the business logic, validating calculation results, approving target thresholds, and driving operational performance based on metric insights.
- **Data Owner:** Accountable for technical lineage, data completeness, source-to-target mapping, ETL data quality checks, and resolving data ingestion or availability issues.
- **Escalation Path:** In case of data quality discrepancies or alignment disputes, issues are escalated to the Data Governance Committee and KPI Advisory Board for review and reconciliation.

Ownership and Governance Policy Matrix

Metric Name	Business Owner	Data Owner	Governance Policy Notes
Net Sales Realization	VP of Sales	Sales Operations Manager	Business Owner (VP of Sales): Reviews variance and coordinates corrective actions. Data Owner (Sales Operations Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Production Throughput	Head of Manufacturing	Production Planning Manager	Business Owner (Head of Manufacturing): Reviews variance and coordinates corrective actions. Data Owner (Production Planning Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Inventory Days	Head of Supply Chain	Inventory Control Manager	Business Owner (Head of Supply Chain): Reviews variance and coordinates corrective actions. Data Owner (Inventory Control Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Forecast Accuracy %	Head of Demand Planning	Demand Planning Analyst	Business Owner (Head of Demand Planning): Reviews variance and coordinates corrective actions. Data Owner (Demand Planning Analyst): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Days Sales Outstanding	CFO / Head of Finance	Accounts Receivable Manager	Business Owner (CFO / Head of Finance): Reviews variance and coordinates corrective actions. Data Owner (Accounts Receivable Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.

EBITDA Margin %	CFO / Head of Finance	Financial Controller	Business Owner (CFO / Head of Finance): Reviews variance and coordinates corrective actions. Data Owner (Financial Controller): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
First Pass Yield %	Head of Quality	Quality Assurance Manager	Business Owner (Head of Quality): Reviews variance and coordinates corrective actions. Data Owner (Quality Assurance Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
OTIF %	Head of Logistics	Logistics Coordinator	Business Owner (Head of Logistics): Reviews variance and coordinates corrective actions. Data Owner (Logistics Coordinator): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Safety Incident Rate	Head of EHS (Environment, Health, and Safety)	EHS Manager	Business Owner (Head of EHS (Environment, Health, and Safety)): Reviews variance and coordinates corrective actions. Data Owner (EHS Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.
Customer Complaint Rate	Head of Customer Service	Customer Service Manager	Business Owner (Head of Customer Service): Reviews variance and coordinates corrective actions. Data Owner (Customer Service Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.

6. Reporting & Dashboard Requirements

Visualization layout and threshold criteria dictate how data is displayed to support decision-making. The matrix below outlines reporting recommendations and performance thresholds for each metric.

Metric Name	Reporting Guidelines	Dashboard Placement	Threshold Guidance
Net Sales Realization	Visual layout: Gauge card and trend line showing actuals vs target of 95-98%.	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $<$ 95%, Amber 95-97%, Green $\geq$ 97%).
Production Throughput	Visual layout: Gauge card and trend line showing actuals vs target of Target based on historical performance and capacity..	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $<$ 80% of target, Amber 80-90% of target, Green $\geq$ 90% of target).
Inventory Days	Visual layout: Gauge card and trend line showing actuals vs target of 30-60 days (specific to product type).	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $>$ 70 days, Amber 60-70 days, Green $\leq$ 60 days).
Forecast Accuracy %	Visual layout: Gauge card and trend line showing actuals vs target of 85-95%.	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $<$ 85%, Amber 85-90%, Green $\geq$ 90%).
Days Sales Outstanding	Visual layout: Gauge card and trend line showing actuals vs target of 45-60 days.	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $>$ 65 days, Amber 60-65 days, Green $\leq$ 60 days).
EBITDA Margin %	Visual layout: Gauge card and trend line showing actuals vs target of 15-25% (industry benchmark dependent).	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $<$ 15%, Amber 15-18%, Green $\geq$ 18%).
First Pass Yield %	Visual layout: Gauge card and trend line showing actuals vs target of 90-98%.	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $<$ 90%, Amber 90-95%, Green $\geq$ 95%).
OTIF %	Visual layout: Gauge card and trend line showing actuals vs target of 95-98%.	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $<$ 95%, Amber 95-97%, Green $\geq$ 97%).
Safety Incident Rate	Visual layout: Gauge card and trend line showing actuals vs target of $<$ 1.0.	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $>$ 2.0, Amber 1.0-2.0, Green $<$ 1.0).
Customer Complaint Rate	Visual layout: Gauge card and trend line showing actuals vs target of $<$ 50 per 100,000 units.	Include on the Executive Performance Dashboard as a primary card.	Green: $\geq$ Target, Amber: Within 5% of target, Red: Below target (Red $>$ 75, Amber 50-75, Green $<$ 50).

7. Assumptions & Constraints

A clear understanding of business and technical assumptions is critical for successful implementation. The following list represents consolidated baseline assumptions and constraints.

General Architectural Assumptions

- **Data Availability:** Source transactional tables are assumed to be loaded into the central reporting repository on a standard nightly batch cadence.
- **Fiscal Calendar:** Standard calendar year rules are assumed unless otherwise explicitly documented in specific financial indicators.
- **System Uptime:** Target source ERP ledgers are assumed to maintain 99.5% uptime during standard reporting extraction windows.

Metric-Specific Assumptions and Limitations

- **Net Sales Realization:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Production Throughput:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Inventory Days:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Forecast Accuracy %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Days Sales Outstanding:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **EBITDA Margin %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **First Pass Yield %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **OTIF %:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Safety Incident Rate:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.
- **Customer Complaint Rate:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.

8. Implementation Considerations

Transitioning these specifications into functional BI tools requires rigorous testing, change management, and data reconciliation. Standard implementation considerations are outlined below.

Standard Implementation Guidelines

- **Data Reconciliation:** All KPI calculation outcomes must be audited and reconciled against official books of record or audited financial statements.
- **User Acceptance Testing (UAT):** Business owners must perform visual and numeric validation of dashboard mockups prior to production sign-off.
- **Change Management:** Training sessions and clear system documentation are required to support user onboarding and ensure high organizational adoption.

Metric-Specific Technical Considerations

- **Net Sales Realization:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Production Throughput:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Inventory Days:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Forecast Accuracy %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Days Sales Outstanding:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **EBITDA Margin %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **First Pass Yield %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **OTIF %:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Safety Incident Rate:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.
- **Customer Complaint Rate:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.

9. Appendix

KPI Glossary

- **KPI (Key Performance Indicator):** A quantifiable measure used to evaluate the success of an organization or activity in meeting performance objectives.
- **KRA (Key Result Area):** Primary focus areas of outcomes or outputs for which an organizational unit or role is responsible.
- **Numerator:** The upper portion of a division representing the measured subset of occurrences.
- **Denominator:** The lower portion of a division representing the total base population.

Data Quality Principles

- **Accuracy:** Data correctly represents the real-world operational event it records.
- **Completeness:** All necessary dataset components are present without omission.
- **Consistency:** Metrics align across various systems, business units, and report interfaces.
- **Timeliness:** Updates occur within the required reporting cadence and operational windows.

Acronym Reference

- **ERP:** Enterprise Resource Planning
- **SAP FI-CO:** Financial Accounting & Controlling
- **SAP SD:** Sales & Distribution
- **SAP MM:** Materials Management
- **BI / DWH:** Business Intelligence / Data Warehouse
- **UAT:** User Acceptance Testing